

AI Data Quality Analysis Report

Comprehensive Synthetic Data Validation

Dataset Information:

Real Dataset: 2000 rows, 8 columns

Synthetic Dataset: 2000 rows, 8 columns

Analysis Date: 06/01/2026

Executive Summary

The validation analysis reveals a comprehensive assessment of synthetic data generation for a healthcare-related dataset with 2,000 rows and 8 columns. The synthetic data demonstrates an exceptional level of statistical similarity to the original dataset, with near-perfect replication across multiple validation dimensions. The dataset covers key variables including ID, AGE, SEX, BMI, CHILDREN, SMOKER, REGION, and CHARGES.

Key statistical observations include:

- 100% column compatibility between real and synthetic datasets
- Perfect range and distribution matching across all variables
- Zero statistically significant differences in correlation structures
- Complete data completeness (1.0 ratio)
- Consistent data types across all columns

However, despite the seemingly perfect replication, the overall data quality is paradoxically rated as "POOR", suggesting potential underlying issues in the synthetic data generation process that may not be immediately apparent through traditional statistical tests. The exact match rate of 1.0 (2,000/2,000 rows) is flagged as a warning, indicating potential over-fitting or lack of meaningful variation in the synthetic data.

The analysis suggests a need for deeper scrutiny of the synthetic data generation methodology, with a focus on introducing controlled variability while maintaining core statistical properties of the original dataset.

Key Findings

The validation results present a nuanced picture of synthetic data quality:

Statistical Consistency:

- All 36 total tests showed remarkable consistency
- 18 tests passed without any statistical significant differences
- Zero failed tests detected
- Zero critical issues identified

Detailed Breakdown:

1. Range Validation:

- Perfect match in min/max values for all numeric columns
- Identical mean and standard deviation across all variables
- Exact preservation of categorical domains (SEX, SMOKER, REGION)

2. Distribution Tests:

- Kolmogorov-Smirnov (KS) tests showed p-values of 1.0 for all numeric variables
- Chi-square tests for categorical variables returned non-significant results
- All distribution tests resulted in "ACCEPT" status

3. Correlation Structure:

- 100% correlation matrix preservation
- No significant differences detected in inter-variable relationships
- Maintained complex interactions between variables like AGE and CHARGES

4. Privacy and Quality Metrics:

- Perfect data completeness (1.0 ratio)
- Full data type consistency (8/8 columns)
- Nearest Neighbor Distance Ratio and Distance tests returned successful results

Anomalous Observations:

- Exact match rate of 1.0 (2,000/2,000 rows) raises potential over-fitting concerns
- Multivariate energy test encountered a processing error

The findings suggest a synthetic dataset that statistically mirrors the original data with extraordinary precision, yet this perfection itself becomes a point of critical examination.

Statistical Quality

The statistical quality assessment reveals a comprehensive validation approach across multiple dimensions:

Distribution Tests:

- 5 Kolmogorov-Smirnov tests for numeric variables
- 3 Chi-square tests for categorical variables
- All tests returned p-values of 1.0, indicating perfect statistical similarity
- No statistically significant differences detected

Correlation Validation:

- 10 total correlation comparisons
- Zero significant differences identified
- Perfect preservation of inter-variable correlation structures
- Maintained complex relationships like AGE-CHARGES correlation (0.304)

Statistical Tests:

- 5 Welch's t-tests comparing means
- All tests returned statistic of 0 and p-value of 1
- Identical means across real and synthetic datasets
- No statistically significant variations

Multivariate Analysis:

- Total variation distance test: non-significant
- Kullback-Leibler divergence test: non-significant
- Jennrich test for correlation matrices: successful
- Energy test encountered processing error

Outlier Detection:

- Consistent outlier patterns in BMI (11 outliers, 0.55% of data)
- Consistent outlier patterns in CHARGES (191 outliers, 9.55% of data)
- Identical outlier identification in both datasets

Limitations and Concerns:

- Perfect statistical matching might indicate over-fitting
- Lack of controlled variability
- Potential algorithmic bias in synthetic data generation

The statistical quality suggests an algorithmically precise replication that paradoxically raises concerns about the synthetic data's genuine representativeness.

Practical Usefulness

Practical usefulness assessment reveals a complex evaluation of the synthetic dataset's applicability:

Data Utility Dimensions:

- Complete preservation of original data characteristics
- Maintained statistical properties across all variables
- Identical distribution and correlation structures
- Perfect representation of categorical domains

Privacy Preservation:

- Nearest Neighbor Distance Ratio indicates minimal privacy risks

- No identifiable individual-level information leakage
- Consistent anonymization across dataset

Synthetic Data Quality:

- Statistically indistinguishable from original data
- Preserved complex multivariate relationships
- Maintained domain-specific characteristics (healthcare context)

Potential Use Cases:

- Machine learning model training
- Statistical analysis and modeling
- Algorithmic testing and development
- Privacy-preserving data sharing

Limitations:

- Exact 1.0 match rate suggests potential over-fitting
- Lack of controlled variability might limit generalizability
- Potential loss of nuanced real-world complexities

Recommendations for Improvement:

- Introduce controlled noise in synthetic data generation
- Implement more sophisticated anonymization techniques
- Develop adaptive synthetic data generation algorithms

The practical usefulness is high in controlled environments but requires careful interpretation and potential methodological refinements.

Critical Issues

No critical issues were explicitly identified in the validation results. However, several potential concerns warrant careful consideration:

1. Exact Match Rate Anomaly:

- 100% of synthetic rows exactly match real rows
- Potential over-fitting of synthetic data generation algorithm
- Risks losing meaningful variability and representativeness

2. Multivariate Energy Test Error:

- Test encountered processing error
- Indicates potential limitations in deep statistical analysis
- Suggests need for more robust multivariate testing methodologies

3. Statistical Perfection Paradox:

- Extraordinarily perfect statistical matching
- Raises questions about genuine data representation
- Potential algorithmic bias in synthetic data generation

While no critical failures were detected, these nuanced concerns suggest a need for methodological refinement in the synthetic data generation process.

Variable Range and Type Checks

Variable Analysis:

Numeric Variables:

1. ID:

- Range: 20 to 99,966
- Type: Consistent (integer)
- No anomalies detected

2. AGE:

- Range: 12.96 to 70.31
- Type: Consistent (floating-point)
- Reasonable age distribution

3. BMI:

- Range: 15.26 to 53.95
- Type: Consistent (floating-point)
- Outliers detected (11 instances)

4. CHILDREN:

- Range: 0 to 5
- Type: Consistent (integer)
- No anomalies detected

5. CHARGES:

- Range: -522.35 to 54,680.38
- Type: Consistent (floating-point)
- Outliers detected (191 instances)

Categorical Variables:

1. SEX:

- Values: "female", "male"
- Type: Consistent
- No anomalies detected

2. SMOKER:

- Values: "yes", "no"
- Type: Consistent
- No anomalies detected

3. REGION:

- Values: "southeast", "northwest", "northeast", "southwest"
- Type: Consistent
- No anomalies detected

No significant type mismatches or unreasonable ranges were identified.

Real Data Quality by Variable

Detailed Variable Assessment:

1. ID:

- Range: 20 to 99,966
- Mean: 50,238.75
- Plausible for unique identifier
- No unreasonable values detected

2. AGE:

- Range: 12.96 to 70.31
- Mean: 39.69
- Slightly low minimum age (12.96 years)
- Maximum age reasonable

- Distribution appears realistic

3. BMI:

- Range: 15.26 to 53.95
- Mean: 30.24
- 11 outliers detected (>45 BMI)
- Reflects realistic body weight variations
- Consistent with healthcare population data

4. CHILDREN:

- Range: 0 to 5
- Mean: 1.27
- Reasonable family size distribution
- No unreasonable values

5. CHARGES:

- Range: -522.35 to 54,680.38
- Mean: 13,253.52
- Negative charge value unusual
- High variability in medical charges
- 191 outliers detected
- Reflects complex healthcare billing landscape

Categorical Variables:

- SEX: Binary classification
- SMOKER: Binary classification
- REGION: Four distinct categories

Overall, the real data demonstrates realistic variations with some expected outliers in medical and health-related variables.

Actionable Recommendations

Recommendations for Synthetic Data Generation Improvement:

1. Controlled Variability Enhancement

- Implement adaptive noise injection techniques

- Develop probabilistic variation mechanisms
- Create controlled randomness without losing core statistical properties
- Use techniques like differential privacy to introduce subtle variations

Justification: The current 100% match rate suggests over-fitting, potentially limiting the synthetic data's generalizability and representativeness.

2. Advanced Anonymization Strategies

- Develop more sophisticated privacy preservation algorithms
- Implement multi-layered anonymization techniques
- Ensure minimal information leakage
- Create nuanced privacy protection without losing data utility

Justification: While current privacy metrics appear strong, the exact matching raises potential re-identification risks.

3. Multivariate Distribution Refinement

- Enhance multivariate statistical modeling
- Develop more complex generative algorithms
- Capture higher-order statistical interactions
- Implement machine learning-based generative models

Justification: The current generation process, while statistically precise, might miss complex underlying data generation mechanisms.

4. Outlier and Extreme Value Handling

- Develop sophisticated outlier generation strategies
- Create probabilistic models for extreme value generation
- Maintain statistical properties while introducing realistic variations
- Implement domain-specific constraint modeling

Justification: Current outlier handling appears uniform across variables, potentially losing nuanced real-world complexity.

5. Continuous Validation Framework

- Develop a comprehensive, iterative validation methodology
- Create dynamic testing protocols
- Implement machine learning-based validation metrics

- Establish continuous improvement mechanisms

Justification: The current validation approach, while comprehensive, could benefit from more adaptive and learning-based validation techniques.

Risk Level

Risk Level: MEDIUM

Justification:

- Perfect statistical matching raises potential over-fitting concerns
- 100% exact match rate indicates possible data generation limitations
- No critical failures detected
- Strong privacy preservation metrics
- Potential loss of nuanced real-world variability

The medium risk level stems from the synthetic data's extraordinary statistical precision, which paradoxically introduces uncertainties about its genuine representativeness and generalizability.

Report Metadata

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